BorNEO HackWknd 2026

GROUP NAME	GEN NEW BEE
TITLE OF THE PROJECT	AI-Powered Robo-Advisor for Students
GROUP LEADER	VINCENT LEE CHENG HONG
TEAM MEMBERS 1	PIONG JIA JIN
TEAM MEMBERS 2	VINCENT SIM ZENG KAI
TEAM MEMBERS 3	GOH MIN JIE
TEAM MEMBERS 4	VANNESS VOON SOON TONG

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BorNEO HackWknd 2026 – Track B Project Report

Project Repository Link: https://github.com/vincentlee001804/roboadvisor

Live Demo: https://roboadvisor-bjfs.onrender.com

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1. Team Information

- **Team Name: Gen New Bee**
- **Contact Details:**
  - Email: bcs24020018@student.uts.edu.my
  - GitHub: https://github.com/vincentlee001804
- **Member Names and Roles:**
  - **[VINCENT LEE CHENG HONG] (Project Lead):** Developed Flask backend, Firebase integration, and AI service integration. Designed UI/UX, mobile optimization, and deployed the application to Render. Computer Science undergraduate.
  - **[PIONG JIA JIN] (Frontend/UI Engineer):** Developed the dashboard UI, image upload functionality, and integrated Chart.js for visual data representation. Computer Science undergraduate.
  - **[VINCENT SIM ZENG KAI] (OCR & Data Engineer):** Managed image processing pipelines and data extraction utilizing the Gemini Vision API. Computer Science undergraduate.
  - **[GOH MIN JIE] (AI Integration Specialist):** Handled LLM prompts, context windows, and API integration for generating personalized financial advice. Computer Science undergraduate.
  - **[VANNESS VOON SOON TONG] (Database & PM):** Designed the Firestore database schema, ensured smooth integration across all application components, and managed the pitch preparation. Computer Science undergraduate.

2. Project Overview

- **Title of the Project:** AI-Powered Robo-Advisor for Students
- **Problem Statement:** Students struggle to manually track expenses, leading to poor financial awareness and overspending.
    - Many students fall into Buy Now Pay Later (BNPL) debt traps due to a lack of awareness about hidden costs, high APRs, and payment obligations.
    - Students lack access to personalized financial advice tailored to their spending patterns and budget constraints.
    - Existing financial management apps are too complex or not designed for student lifestyles and limited budgets.
    - Physical receipts are easily lost, making expense tracking tedious and error-prone.
- **Target Users and Beneficiaries:**
    - **Primary Users:** University and college students managing limited budgets, international students navigating new financial systems, students new to financial independence, and students using BNPL services (GrabPay Later, Atome, etc.).
    - **Beneficiaries:** Students gain improved financial health and debt prevention; educational institutions can enhance student wellness programs; financial institutions get better-informed young customers with reduced default rates; and society benefits from a financially responsible future generation.
- **Objectives of the Solution:**
    - Automate expense tracking by eliminating manual data entry through AI-powered OCR.
    - Provide personalized financial advice by analysing spending patterns to deliver actionable, contextual recommendations.
    - Prevent BNPL debt traps by warning students about hidden costs and high APRs before they commit.
    - Simplify budget management with visual progress indicators and real-time spending updates.
    - Improve financial literacy through proactive advice and insights.
    - Create a frictionless, mobile-first experience.

3. Solution Description

- **Core Features and Functionalities:**
  - **Frictionless Expense Tracking:** Users can upload receipt photos or e-wallet screenshots via a mobile camera. The system allows for manual entry, post-upload item editing, and full expense history viewing with filters.
  - **AI-Powered OCR:** Utilizes Google Gemini Vision API to extract multiple items from receipts, automatically categorize expenses, and handle various receipt formats with high accuracy.
  - **Proactive Financial Advice:** AI analyses spending patterns to provide personalized, actionable recommendations based on the user's budget, complete with intelligent caching for performance.
  - **BNPL Trap Warning System:** Analyses Buy Now Pay Later offers to calculate total cost, effective APR, and risk level, providing personalized warnings to help students make informed decisions.
  - **Budget Management:** Allows setting total monthly budgets and optional category-specific budgets, featuring visual progress bars and real-time spending tracking.
  - **Mobile-Optimized Interface:** Features a responsive design, mobile-first navigation (hamburger menu), touch-friendly buttons, and a modern layout.
- **Technical Architecture:**
  - **Frontend:** Built with Flask utilizing the Jinja2 templating engine, Tailwind CSS for responsive styling, and Vanilla JavaScript for interactivity.
  - **Backend:** Powered by Flask (Python 3.11) featuring RESTful API endpoints, session management, and secure file upload handling.
  - **Database & Storage:** Employs Firebase Firestore for real-time NoSQL data synchronization (users, expenses, budgets collections) and Firebase Storage for automated receipt image handling.
  - **AI & Deployment:** Integrates the Google Gemini API (gemini-3-flash-preview) for OCR, advice, and BNPL analysis, fully deployed on Render.com using a Gunicorn WSGI server.

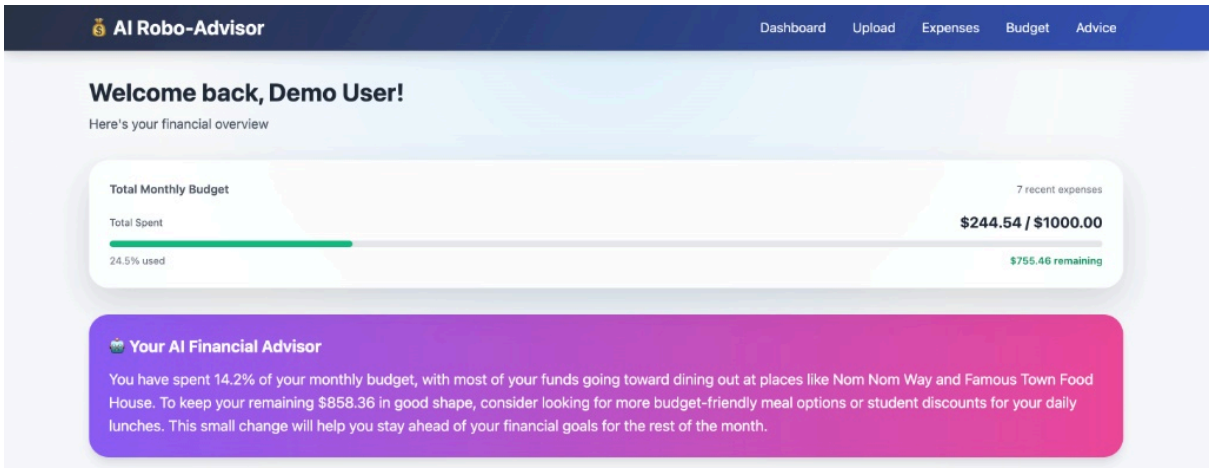
- **Innovation and Uniqueness:**
    - Unlike simple OCR that extracts only totals, our system extracts individual line items from receipts and automatically categorizes each one.
    - Provides proactive, context-aware financial advice that strictly considers the user's specific budget constraints.
    - Features a unique BNPL Risk Analysis tool explicitly targeted to protect students from predatory micro-debts.

4. Implementation Details

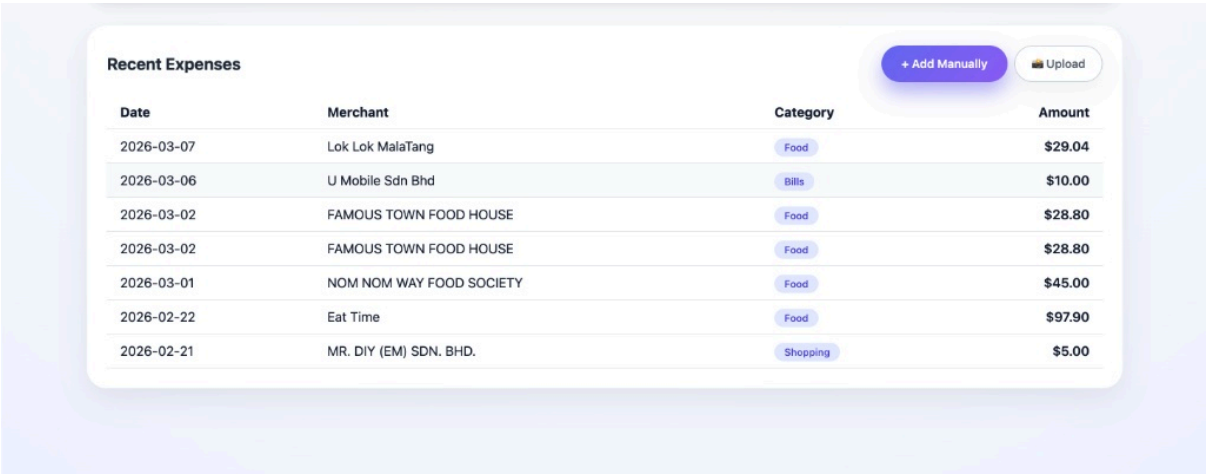
- **Development Process and Workflow:**
  - **Phase 1 (Planning):** Defined core features, designed the Firestore database schema, and selected the Flask/Firebase/Gemini tech stack.
  - **Phase 2 (Backend):** Set up the Flask application, integrated Firebase for database and storage, and implemented CRUD operations and session management.
  - **Phase 3 (AI Integration):** Integrated the Gemini API, developed OCR prompt engineering, built the advice caching system, and implemented fallback mechanisms for API reliability.
  - **Phase 4 (Frontend):** Designed a responsive UI with Tailwind CSS, implemented card-based mobile layouts, and added interactive modals.
  - **Phase 5 (Testing & Refinement):** Tested features locally, optimized touch targets for mobile, and standardized UI consistency.
  - **Phase 6 (Deployment):** Configured Render deployment, set up environment variables, and resolved Python version compatibility issues.
- **Challenges Faced and How They Were Solved:**
  - **Gemini API Compatibility:** The initial model returned 404 errors on Render. We solved this by switching to gemini-3-flash-preview and implementing a fallback chain (gemini-pro → gemini-1.5-pro → gemini-1.5-flash).
  - **Firebase Deployment:** Render restricts service account JSON uploads. We implemented environment variable support (FIREBASE_SERVICE_ACCOUNT) that falls back to a file for local development.
  - **AI Advice Truncation:** Responses were incomplete. We removed the max_output_tokens limit, improved prompt engineering for conciseness, and simplified the parsing logic.
  - **Python Versioning:** Encountered Pillow and protobuf issues on Render. We fixed this by creating a runtime.txt specifying Python 3.11.9 and updating dependencies.

5. Prototype Showcase

- **Navigation & Identity**
  - **Header:** Features the "AI Robo-Advisor" logo with links to the Dashboard, Upload, Expenses, Budget, and Advice sections.
  - **User Greeting:** A personalized "Welcome back, Demo User!" message provides immediate context for the current session.



- **Financial Overview (Progress Bar)**
  - **Total Monthly Budget:** Set at $1,000.00.
  - **Total Spent:** $244.54 (indicated by a green progress bar showing 24.5% used).
  - **Remaining Funds:** Clearly labeled as $755.46 remaining to help the user gauge their "safe to spend" amount for the rest of the month.

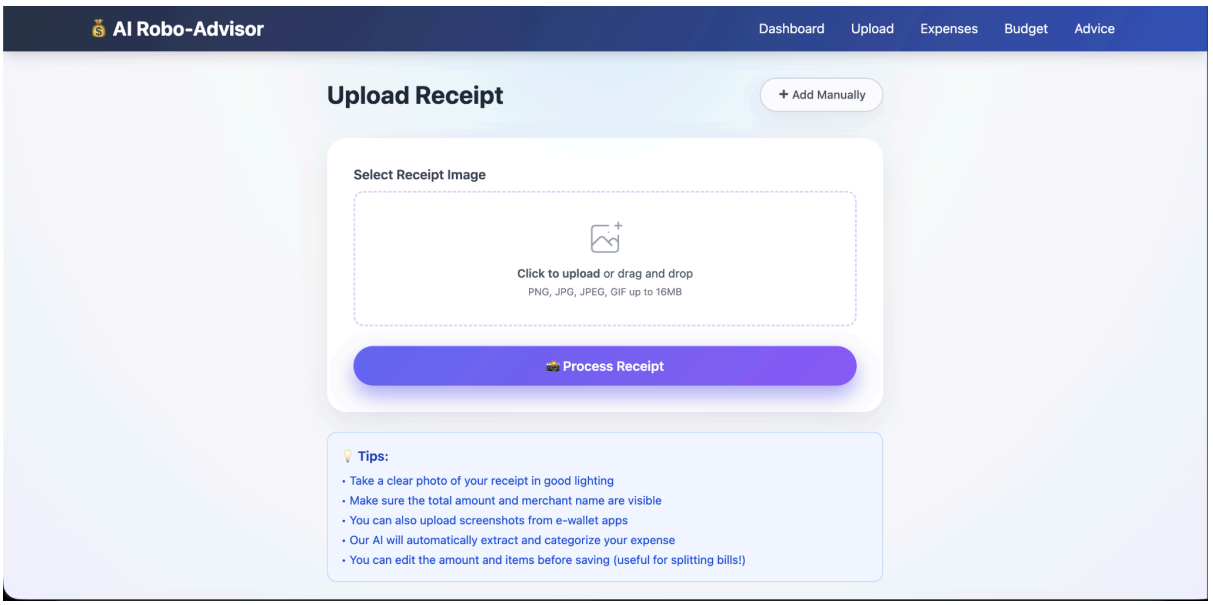


Recent Expenses

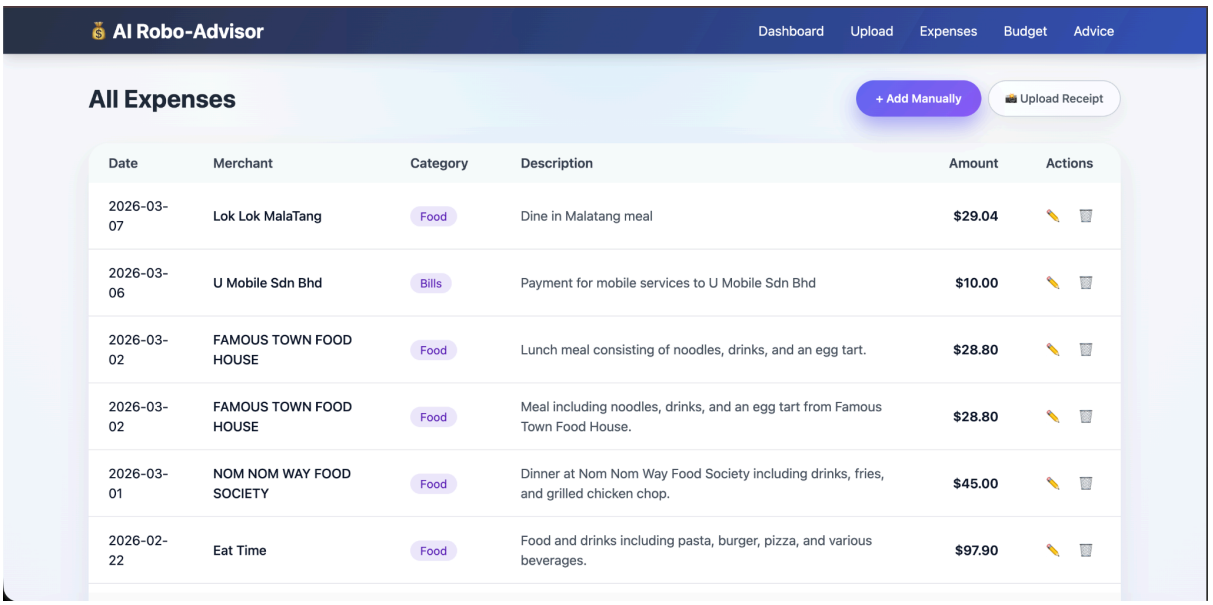
+ Add Manually | Upload

Date	Merchant	Category	Amount
2026-03-07	Lok Lok MalaTang	Food	$29.04
2026-03-06	U Mobile Sdn Bhd	Bills	$10.00
2026-03-02	FAMOUS TOWN FOOD HOUSE	Food	$28.80
2026-03-02	FAMOUS TOWN FOOD HOUSE	Food	$28.80
2026-03-01	NOM NOM WAY FOOD SOCIETY	Food	$45.00
2026-02-22	Eat Time	Food	$97.90
2026-02-21	MR. DIY (EM) SDN. BHD.	Shopping	$5.00

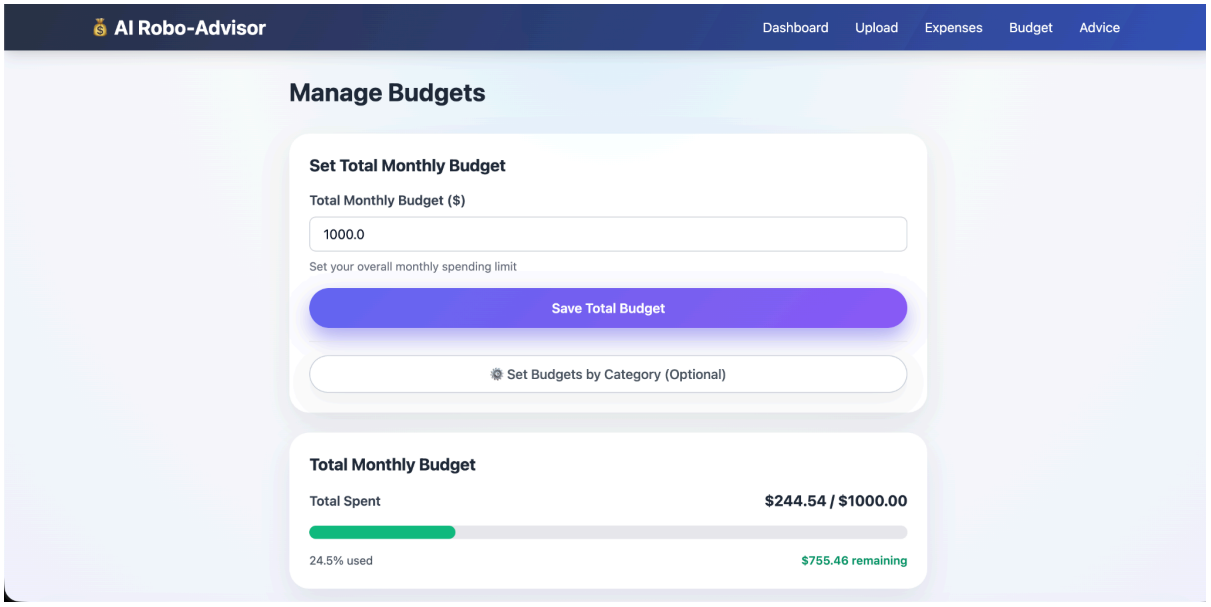
- Core Upload Functionality
    - **Drag and Drop Zone:** A large, dashed-border area where users can click to open a file browser or simply drag a receipt image. It specifies supported formats (PNG, JPG, JPEG, GIF) and a size limit of 16MB.
    - **"Process Receipt" Button:** A prominent purple button that initiates the AI analysis once an image is selected.
    - **"Add Manually" Button:** Located at the top right of the card, serving as a fallback for users who don't have a receipt or prefer to type in details themselves.



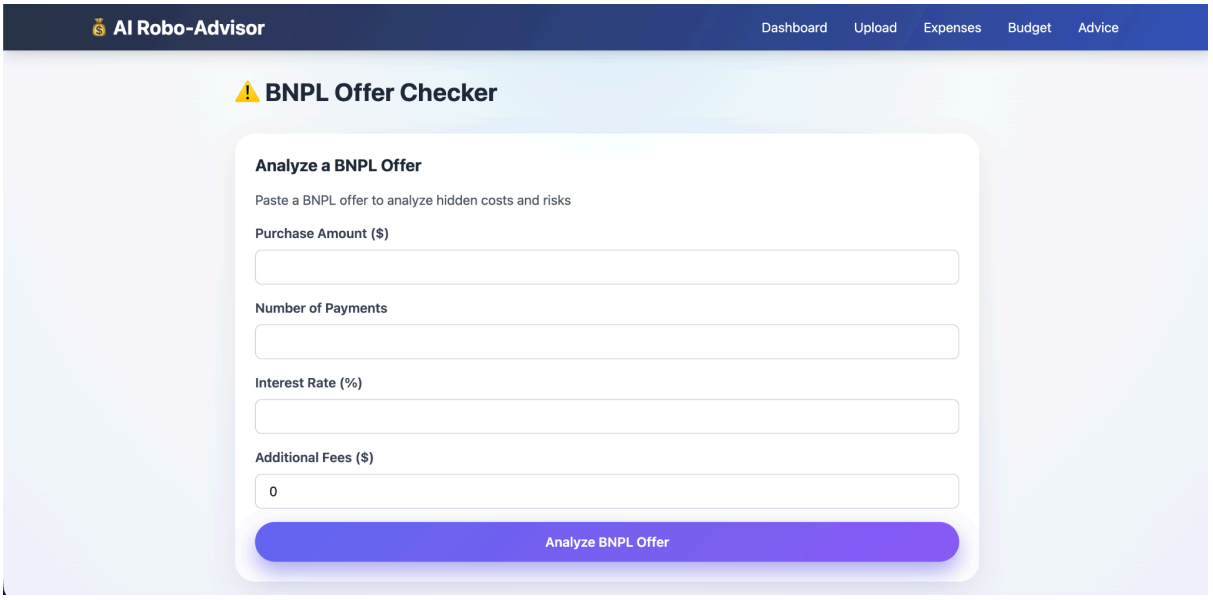
- **Data Visualization (The Expenses Table)**
    - **Date:** Shows when the expense occurred, arranged in reverse chronological order (latest first).
    - **Merchant:** Identifies the business or service provider (e.g., Lok Lok MalaTang, U Mobile Sdn Bhd).
    - **Category:** Features color-coded badges like **Food** (purple) and **Bills** (lavender) to help users quickly identify spending types.
    - **Description:** Provides specific details about the purchase, such as "Lunch meal consisting of noodles, drinks, and an egg tart."
    - **Amount:** Displays the total cost of the transaction in bold text for readability.
    - **Actions:** Offers interactive icons for each entry—a **pencil icon** for editing details and a **trash can icon** for deleting the record.



- **Budget Configuration**
  - **Set Monthly Limit**: Includes a numerical input field (currently **$1000.0**) and a purple "Save Total Budget" button to define spending goals.
  - **Detailed Control**: An optional button allows users to set specific budgets by category, such as food or bills.
  - **Visual Tracking**: A green progress bar and percentage tracker show that **24.5%** ($244.54) of the budget has been used.
  - **Financial Status**: Clearly displays the remaining balance of **$755.46** to help users stay within their limits.



- **BNPL Analysis Tool**
    - **Purchase Amount ($):** A field to enter the total cost of the item or service.
    - **Number of Payments:** A field to specify the installment frequency or total number of scheduled payments.
    - **Interest Rate (%):** An input for any applicable annual or monthly interest rates.
    - **Additional Fees ($):** A field (defaulted to **0**) to account for processing, late, or administrative fees.
    - **Analyze BNPL Offer Button:** A prominent purple button that processes the data to reveal hidden costs.



6. Impact & Scalability

- **Potential Real-World Applications:** Can be integrated into university student portals and financial aid offices.
    - Financial institutions can offer the tool to student account holders as part of a wellness program.
    - Can function as a standalone mobile app or integrate with existing e-wallets like GrabPay and Touch 'n Go.
- **How the Solution Can be Scaled or Improved:**
    - **Short-term:** Add Firebase Authentication for multi-user support, implement CSV/PDF data exports, and add predictive spending analytics charts.
    - **Medium-term:** Develop native iOS and Android apps, integrate with e-wallet APIs, and add bill reminders.
    - **Long-term:** Utilize machine learning for spending prediction, support multiple currencies for international students, and provide a white-label solution for institutions.
- **Sustainability Considerations:**
    - The cloud infrastructure utilizes generous free tiers (Firebase, Gemini API, Render) making it highly cost-effective to run.
    - The digital-first nature reduces the need for paper receipts, minimizing environmental impact.
    - Future business models could include a freemium structure, API licensing, or sponsorships by financial institutions.

7. Future Work

- **Planned Enhancements:**
  - Implementation of user accounts, profiles, and data isolation via Firebase Authentication.
  - Advanced reporting features including monthly/yearly breakdowns, tax report generation, and predictive spending analysis.
  - Social community features such as anonymized peer comparisons, group budgets, and financial tip sharing.
- **Long-Term Vision:** To become the leading financial management platform for students in Southeast Asia.
  - **Mission:** Empower students with AI-driven financial intelligence to build a financially responsible generation.
  - **Target Impact:** Serve 1 million+ students across Southeast Asia within 3 years, reduce student debt by 20% in partner institutions, and improve financial literacy scores by 30%.

8. References

- Flask (Python Web Framework) - Version: 3.0.0. Usage: Web framework, routing, templating.
- Google Gemini API (Model: gemini-3-flash-preview). Usage: OCR, financial advice, BNPL analysis.
- Firebase (Firestore & Storage). Usage: Database and file storage.
- Tailwind CSS - Version: 2.2.19. Usage: UI styling and responsive design.
- Pillow (Python Imaging Library) - Version: 10.3.0+. Usage: Image processing.
- Gunicorn - Version: 21.2.0. Usage: Production WSGI server.
- Firebase Admin SDK - Version: 6.5.0+. Usage: Firebase integration.
- Google Generative AI SDK - Version: 0.3.2. Usage: Gemini API integration.

9. AI Acknowledgement

AI Tools Used:

1. Cursor AI / AI Coding Assistant

- Used for: Code generation, debugging, problem-solving, and development assistance
- Usage: Assisted with Flask backend development, Firebase integration, AI service implementation, UI/UX design, deployment configuration, and troubleshooting
- Examples:
    - Generated Flask routes and template code
    - Helped debug Firebase integration issues
    - Assisted with Gemini API integration and prompt engineering
    - Provided solutions for deployment challenges (Python version compatibility, Pillow build issues)
    - Helped optimize mobile responsiveness and UI consistency

2. Google Gemini API (Product Integration)

- Used for: Core product functionality (OCR, financial advice, BNPL analysis)
- Usage: Integrated as part of the application's AI services
- Note: This is a product feature, not a development tool. The application uses Gemini API to provide OCR and financial advice to end users.

AI Usage Summary:

Development Phase:

- AI coding assistants (Cursor AI) were used extensively for:
- Generating initial code structure and boilerplate
- Debugging errors and finding solutions to technical challenges
- Optimizing code performance and best practices
- Writing documentation and comments
- Solving deployment and compatibility issues

Product Features:

- Google Gemini API is integrated as a core feature of the application:
- OCR (Optical Character Recognition) for receipt processing
- Financial advice generation based on spending patterns
- BNPL (Buy Now Pay Later) offer analysis and risk assessment